# Defensive Security Project

Clarissa Martinez

# Table of Contents

This document contains the following resources:

01

**Monitoring Environment**

02

**Attack Analysis**

03

**Project Summary & Future Mitigations**

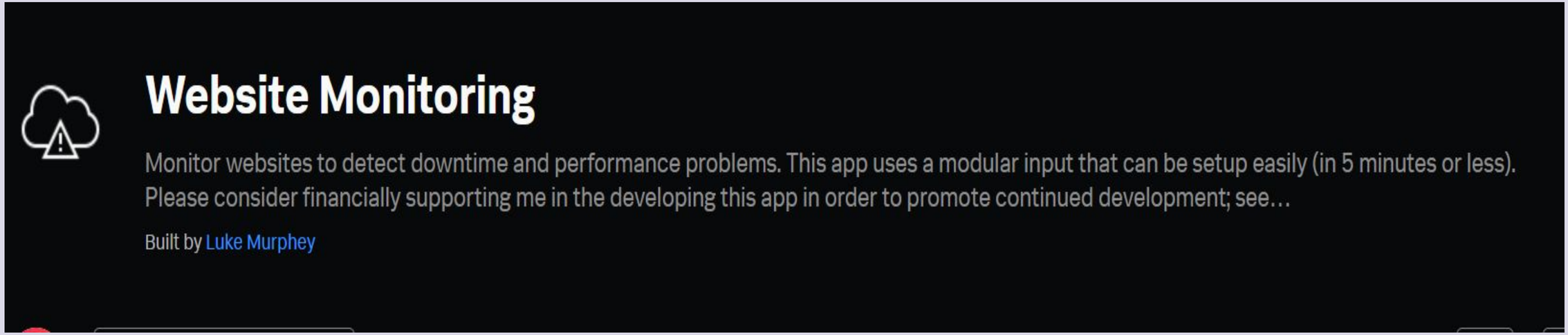
# Monitoring Environment

# Scenario

- For this project we are given the role of a SOC analyst at VSI (Virtual Space Industries). The VSI company specializes in virtual reality programs and rumor is that Jobe Corp may launch a cyber attack to disrupt VSI’s business. Using Splunk we will monitor against potential attacks that might occur within our systems
- The VSI products that will be used to monitor include:
  - An administrative webpage: https://vsi-corporation.azurewebsites.net/
  - Apache web server
  - Windows operating system (runs many of VSI's back-end operations)

# [“Add-On” App]

# [Add-On App Name]



# Website Monitoring (Add-On App)

Benefits of using the Website Monitoring add-on:

-know if website is functioning properly

-malware control

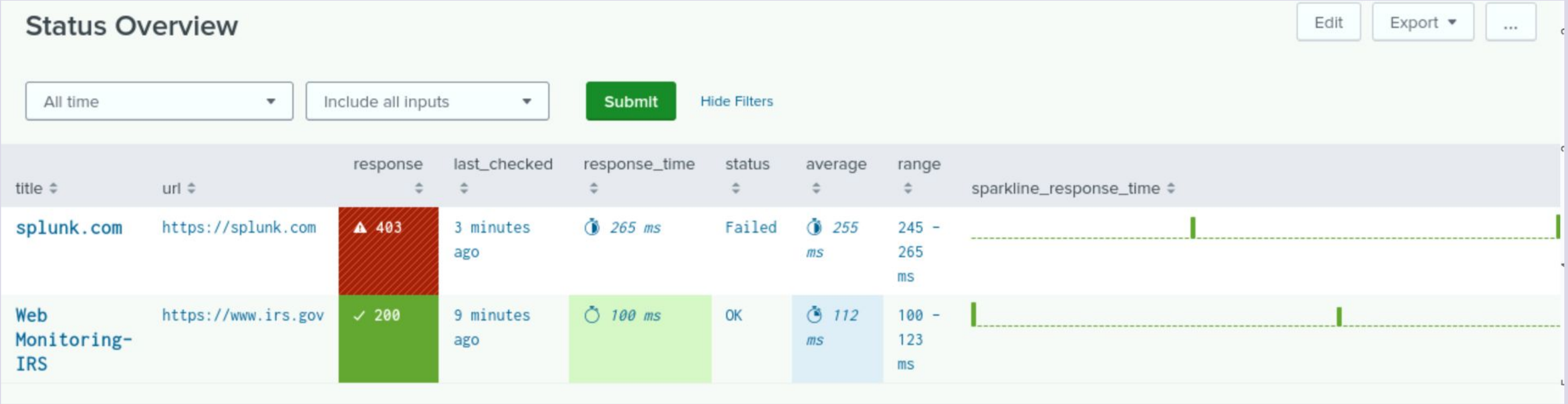
-minimizing the loss of clients and revenue

-continuous discovery and improvement

-optimized website performance

-organizing the company’s technological resources such as bandwidth, access to servers, use of RAM memory, and system speed

# [Add-On App Name]




| title | url | response | last_checked | response_time | status | average | range | sparkline_response_time |
|---|---|---|---|---|---|---|---|---|
| splunk.com | https://splunk.com | 403 | 3 minutes ago | 265 ms | Failed | 255 ms | 245 - 265 ms | |
| Web Monitoring-IRS | https://www.irs.gov | 200 | 9 minutes ago | 100 ms | OK | 112 ms | 100 - 123 ms | |

We monitored 2 sites, splunk.com and irs.gov.

Splunk.com seems to return a 403 error code every ping, meaning that the server recognizes our request but denies it.
IRS.gov is responding as normal.

# Logs Analyzed

## 1 Windows Logs

The Windows logs contain data involving the dates, times, users and how those users gained access into the web servers.

## 2 Apache Logs

The set of Apache logs come from the Apache web servers that host VSI’s web application. The logs demonstrate normal business operations within the company.

# Windows

# Reports

| Report Name | Report Description |
|---|---|
| Users | A table of users, the count of their usage and the percentage of usage. |
| Success & Failure of Windows Activities | A comparison between the success and failure of Windows activities. This will show VSI if there is a suspicious level of failed activities on their server. |
| Severity Levels | Displays the severity levels, the count, and the percentage of each. |
| Signatures & Associated Signature IDs | A table of signatures and associated signature IDs. |

# Images of Reports—Windows

## WSL - Users

Edit ▾ | More Info ▾ | Add to Dashboard

A report with a table of users, the count of their usage and the percentage of usage.

All time ▾

✓ **4,764 events** (before 11/29/23 9:05:07.000 PM)

Job ▾

10 results | 20 per page ▾

| user | count | percent |
|---|---|---|
| user_l | 354 | 7.430730 |
| user_a | 282 | 5.919395 |
| user_m | 275 | 5.772460 |
| user_i | 271 | 5.688497 |
| user_f | 270 | 5.667506 |
| user_h | 269 | 5.646516 |
| user_e | 269 | 5.646516 |
| user_c | 267 | 5.604534 |
| user_d | 264 | 5.541562 |
| user_b | 263 | 5.520571 |

# Images of Reports—Windows

## WSL - Success & Failure of Windows Activities

Edit ▾ | More Info ▾ | Add to Dashboard

A report that provides a comparison between the success and failure of Windows activities. This will show VSI if there is a suspicious level of failed activities on their server.

All time ▾

✓ **4,764 events** (1/28/20 9:15:14.000 AM to 11/29/23 8:16:21.000 PM)

Job ▾

2 results | 20 per page ▾

| status | count | percent |
|---|---|---|
| success | 4622 | 97.019312 |
| failure | 142 | 2.980688 |

# Images of Reports—Windows

## WSL - Severity

Edit ▾ | More Info ▾ | Add to Dashboard

A report that displays the severity levels, the count and the percentage of each.

All time ▾

✓ **4,764 events** (1/28/20 9:15:14.000 AM to 11/29/23 8:17:25.000 PM)

Job ▾

2 results | 20 per page ▾

| severity | count | percent |
|---|---|---|
| informational | 4435 | 93.094039 |
| high | 329 | 6.905961 |

# Images of Reports—Windows

## WSL - Signatures & Associated Signature IDs

Edit ▾ | More Info ▾ | Add to Dashboard

A report with a table of signatures and associated signature IDs.

All time ▾

✓ **15 events** (before 11/29/23 8:45:18.000 PM)

Job ▾

15 results | 20 per page ▾

| signature | signature_id |
|---|---|
| A user account was deleted | 4726 |
| A user account was created | 4720 |
| A computer account was deleted | 4743 |
| An account was successfully logged on | 4624 |
| Special privileges assigned to new logon | 4672 |
| An attempt was made to reset an accounts password | 4724 |
| System security access was granted to an account | 4717 |
| A privileged service was called | 4673 |
| A logon was attempted using explicit credentials | 4648 |
| A user account was locked out | 4740 |
| Domain Policy was changed | 4739 |
| A user account was changed | 4738 |
| A process has exited | 4689 |
| The audit log was cleared | 1102 |
| System security access was removed from an account | 4718 |

# Alerts—Windows

Designed the following alerts:

| Alert Name | Alert Description | Alert Baseline | Alert Threshold |
|---|---|---|---|
| Failed Windows Activity | Splunk alerts VSI when the amount of failed windows activity breaches the threshold. | 14 | 15 |

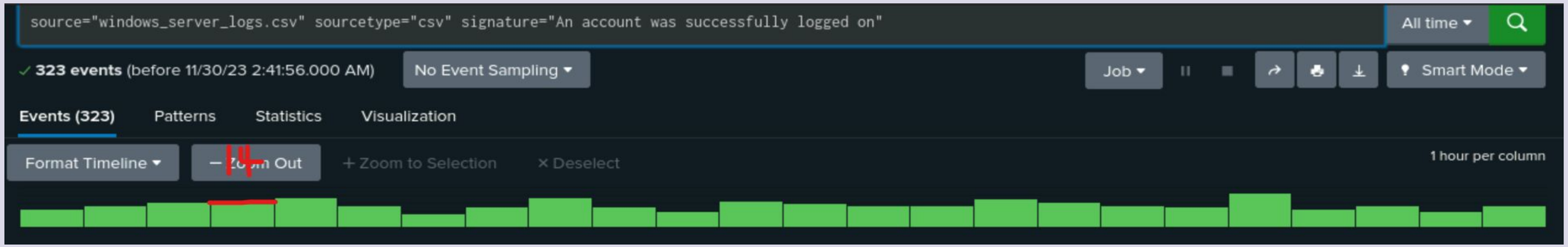
**JUSTIFICATION:**

# Alerts—Windows

Designed the following alerts:

| Alert Name | Alert Description | Alert Baseline | Alert Threshold |
|---|---|---|---|
| Successful Log In Attempt Threshold | Splunk alerts VSI when the amount of log ins breaches the threshold. | 14 | 15 |

**JUSTIFICATION:**



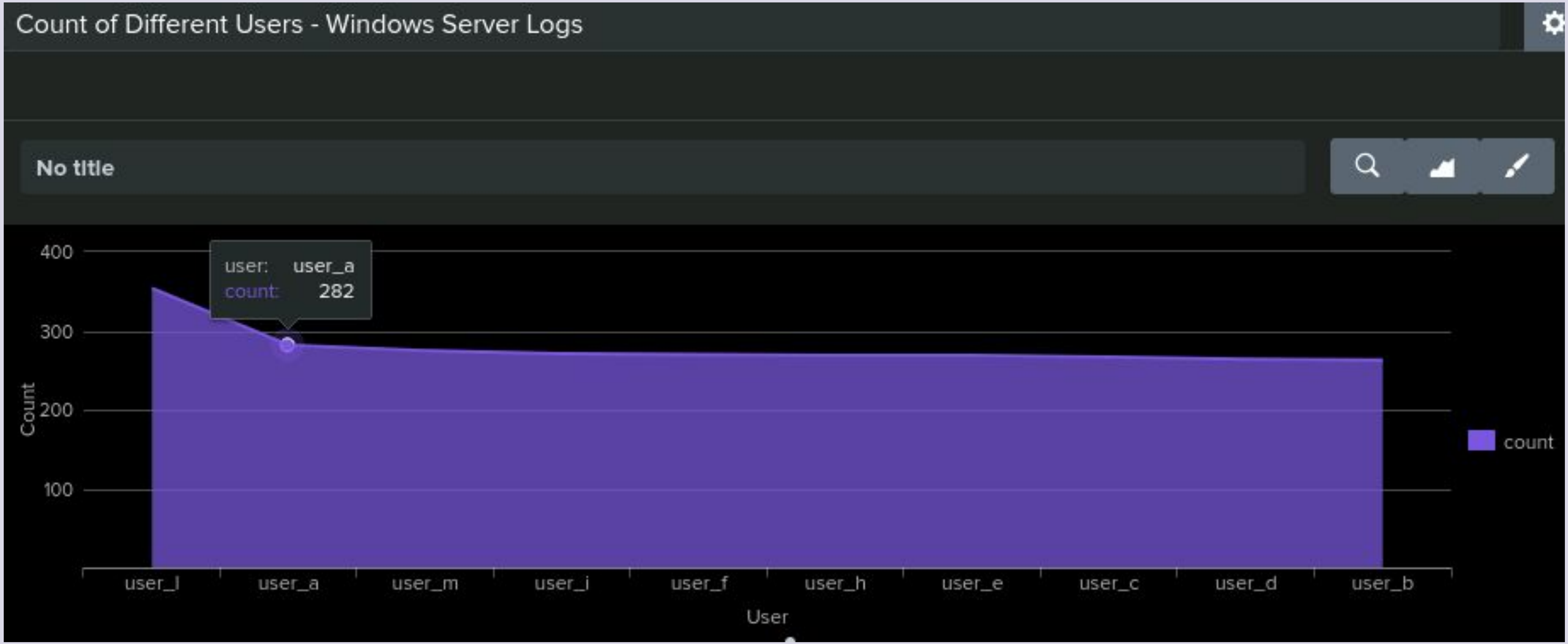
# Alerts—Windows

Designed the following alerts:

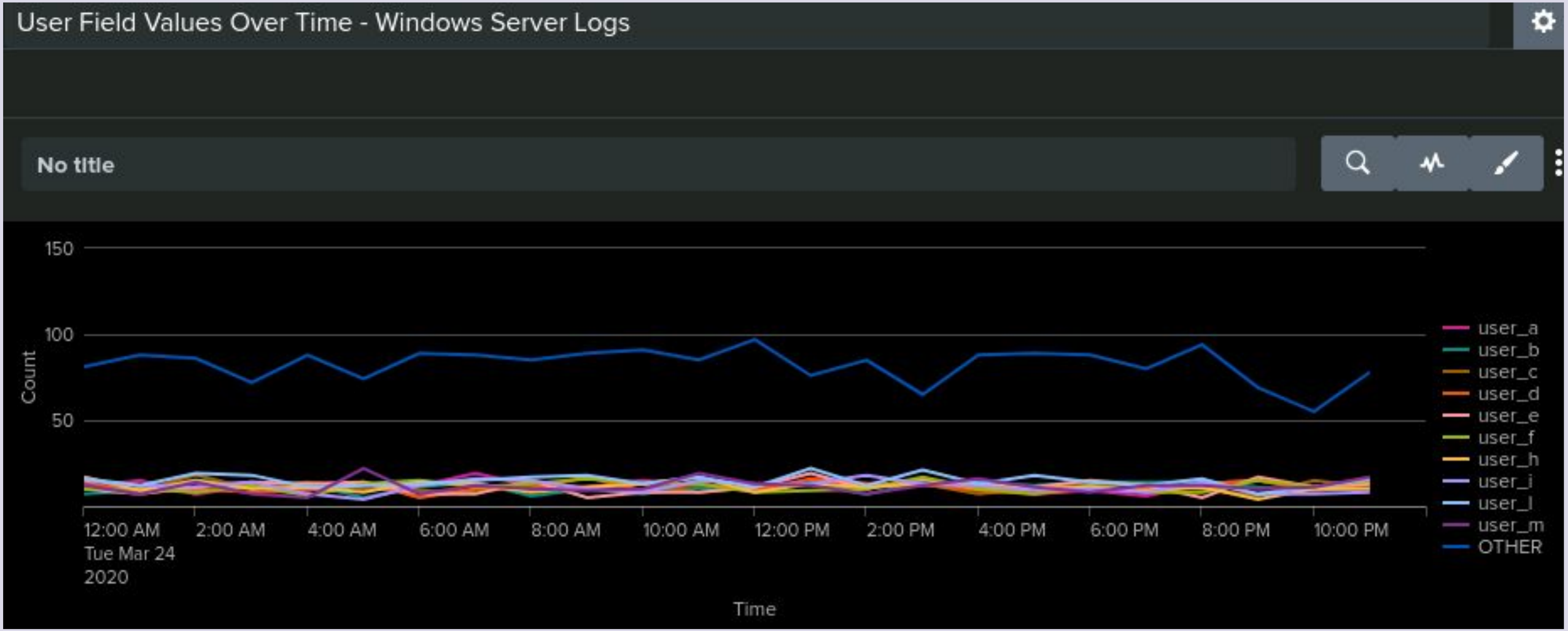
| Alert Name | Alert Description | Alert Baseline | Alert Threshold |
|---|---|---|---|
| Accounts Deleted Threshold | Alert that notifies VSI when an abnormal amount of deleted accounts has occurred. | 12 | 13 |

**JUSTIFICATION:**

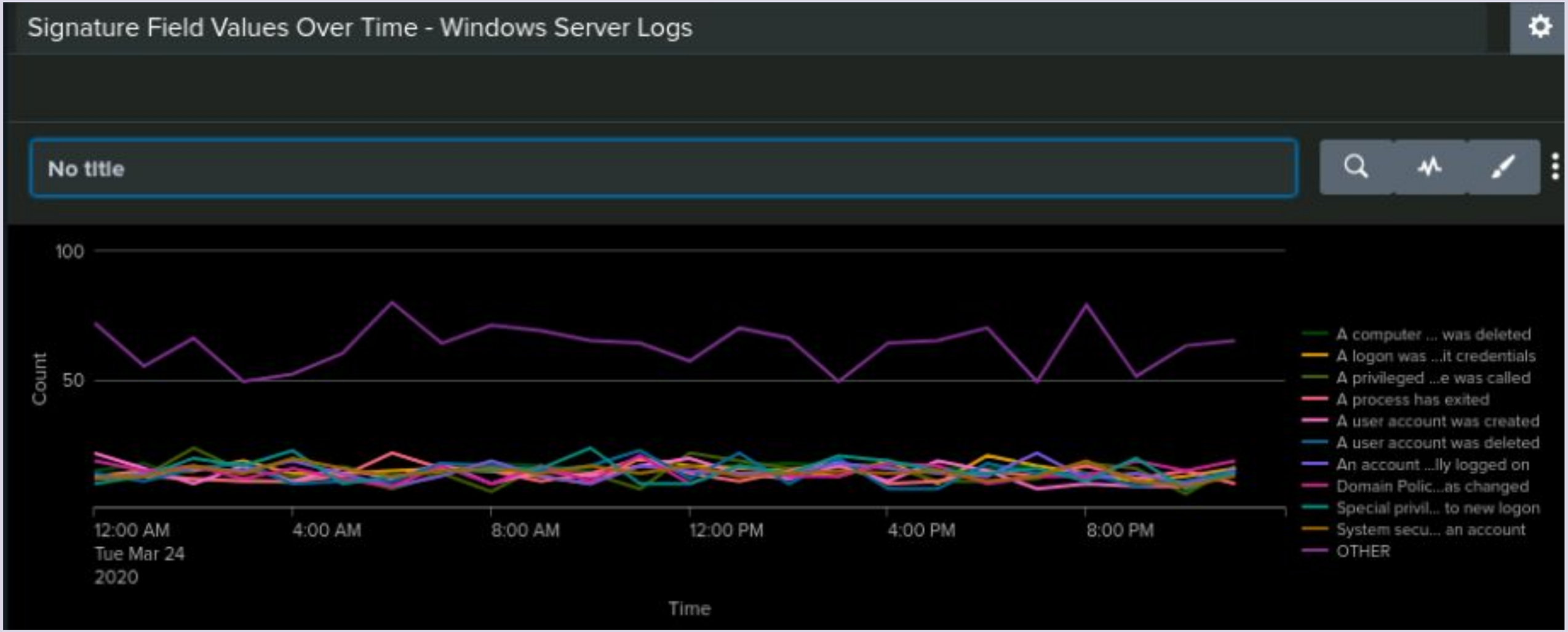
# Dashboards—Windows



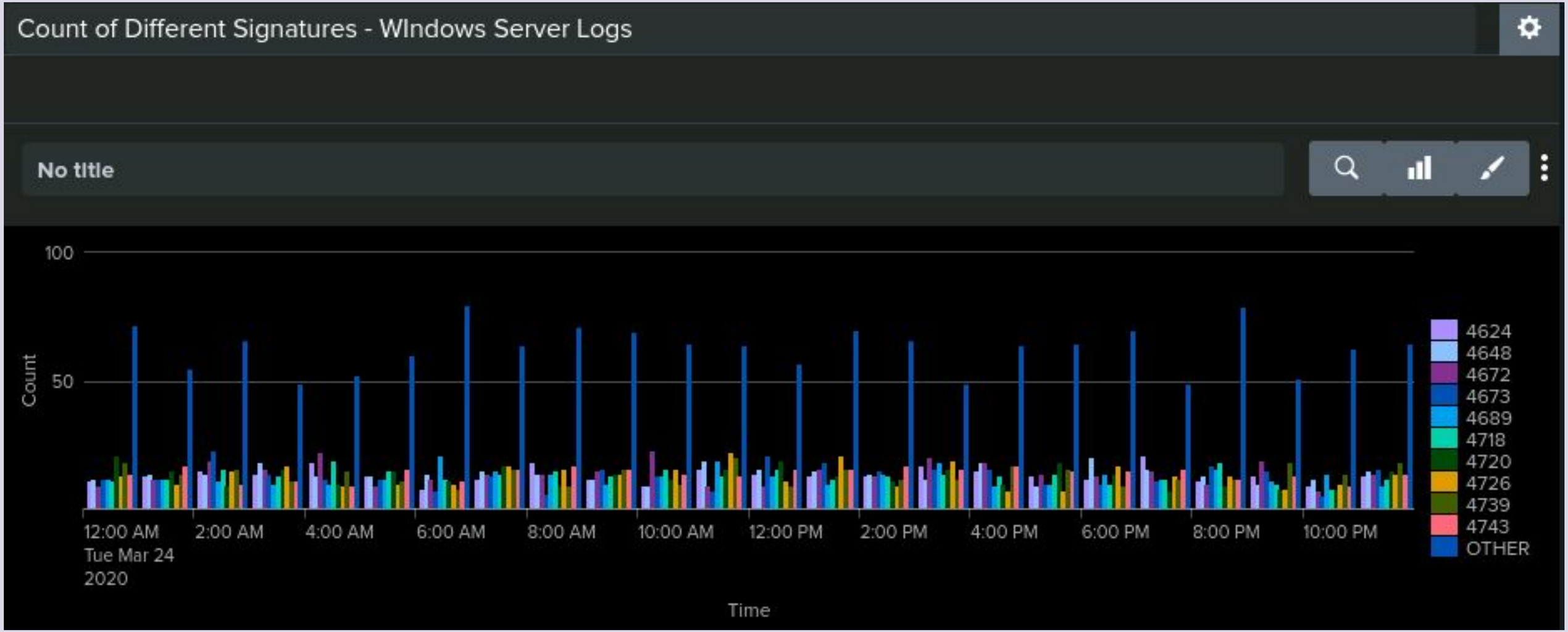
# Dashboards—Windows



# Dashboards—Windows



# Dashboards—Windows



# Apache Logs

# Reports—Apache

Designed the following reports:

| Report Name | Report Description |
|---|---|
| HTTP Methods | A report of different HTTP Methods (GET, POST, HEAD, etc). Provides insight to the type of HTTP activity that is requested against the VSI's web server. |
| Top 10 Referred Domains | A report showing the top 10 domains that refer to VSI's website |
| HTTP Response Code | A report showing the count of each HTTP response code |

# Images of Reports—Apache

## Different HTTP Methods (1)

Edit ▾ | More Info ▾ | Add to Dashboard

A report that shows a table of the different HTTP methods (GET, POST, HEAD, etc).

All time ▾

✓ **10,000 events** (before 11/29/23 5:44:41.000 PM)

Job ▾

4 results | 20 per page ▾

| method | count | percent |
|---|---|---|
| GET | 9851 | 98.510000 |
| POST | 106 | 1.060000 |
| HEAD | 42 | 0.420000 |
| OPTIONS | 1 | 0.010000 |

## Top Domains Referred (1)

Edit ▾ | More Info ▾ | Add to Dashboard

A report that shows the top 10 domains that referred to VSI's website

All time ▾

✓ **10,000 events** (before 11/29/23 5:47:10.000 PM)

Job ▾

10 results | 20 per page ▾

| referer_domain | count | percent |
|---|---|---|
| http://www.semicomplete.com | 3038 | 51.256960 |
| http://semicomplete.com | 2001 | 33.760756 |
| http://www.google.com | 123 | 2.075249 |
| https://www.google.com | 105 | 1.771554 |
| http://stackoverflow.com | 34 | 0.573646 |
| http://www.google.fr | 31 | 0.523030 |
| http://s-chassis.co.nz | 29 | 0.489286 |
| http://logstash.net | 28 | 0.472414 |
| http://www.google.es | 25 | 0.421799 |
| https://www.google.co.uk | 23 | 0.388055 |

# Images of Reports—Apache

## Top Domains Referred (1)

A report that shows the top 10 domains that referred to VSI's website

Edit ▾ | More Info ▾ | Add to Dashboard

All time ▾

✓ **10,000 events** (before 11/29/23 5:47:10.000 PM)

Job ▾

10 results | 20 per page ▾

| referer_domain | count | percent |
|---|---|---|
| http://www.semicomplete.com | 3038 | 51.256960 |
| http://semicomplete.com | 2001 | 33.760756 |
| http://www.google.com | 123 | 2.075249 |
| https://www.google.com | 105 | 1.771554 |
| http://stackoverflow.com | 34 | 0.573646 |
| http://www.google.fr | 31 | 0.523030 |
| http://s-chassis.co.nz | 29 | 0.489286 |
| http://logstash.net | 28 | 0.472414 |
| http://www.google.es | 25 | 0.421799 |
| https://www.google.co.uk | 23 | 0.388055 |

## HTTP Response Codes (1)

A report that shows the count of each HTTP response code.

Edit ▾ | More Info ▾ | Add to Dashboard

All time ▾

✓ **10,000 events** (before 11/29/23 5:48:54.000 PM)

Job ▾

8 results | 20 per page ▾

| status | count | percent |
|---|---|---|
| 200 | 9126 | 91.260000 |
| 304 | 445 | 4.450000 |
| 404 | 213 | 2.130000 |
| 301 | 164 | 1.640000 |
| 206 | 45 | 0.450000 |
| 500 | 3 | 0.030000 |
| 416 | 2 | 0.020000 |
| 403 | 2 | 0.020000 |

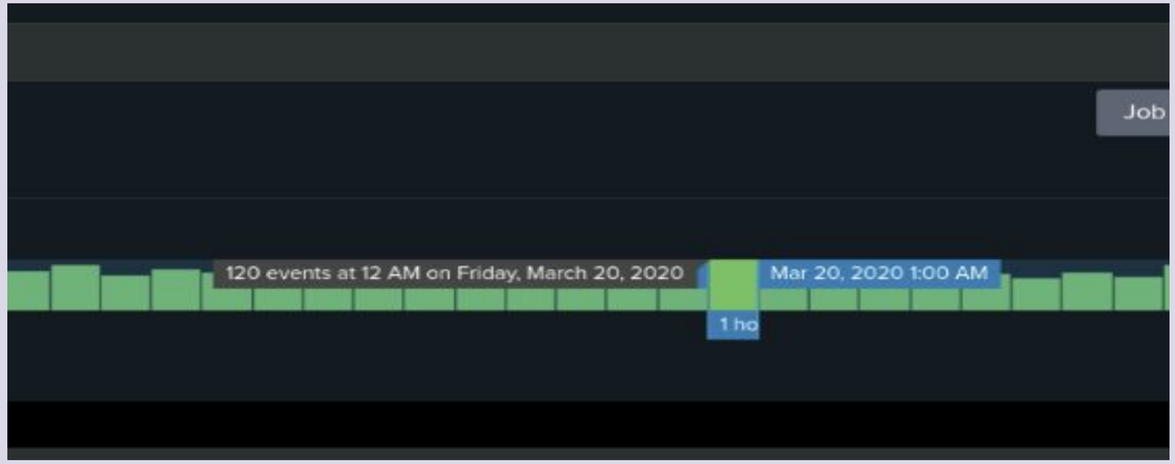
# Alerts—Apache

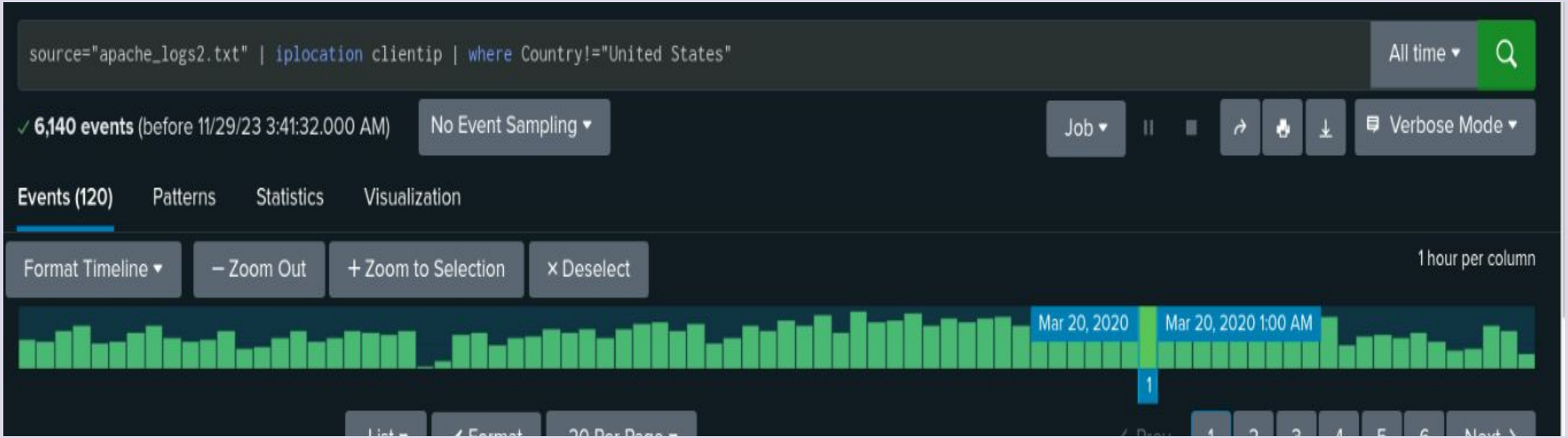
Designed the following alerts:

| Alert Name | Alert Description | Alert Baseline | Alert Threshold |
|---|---|---|---|
| Non-US Activity (Hourly) | The hourly activity from any country besides the United States | 90 | 100 |

**JUSTIFICATION: The standard number of logs per hour always seemed to range between 70-90. However a few events exceeded over 100 with the most number of events showing 120. Seeing a greater number of events from the threshold should be checked on.**

# Alerts—Apache (Non-US Activity Hourly) Image



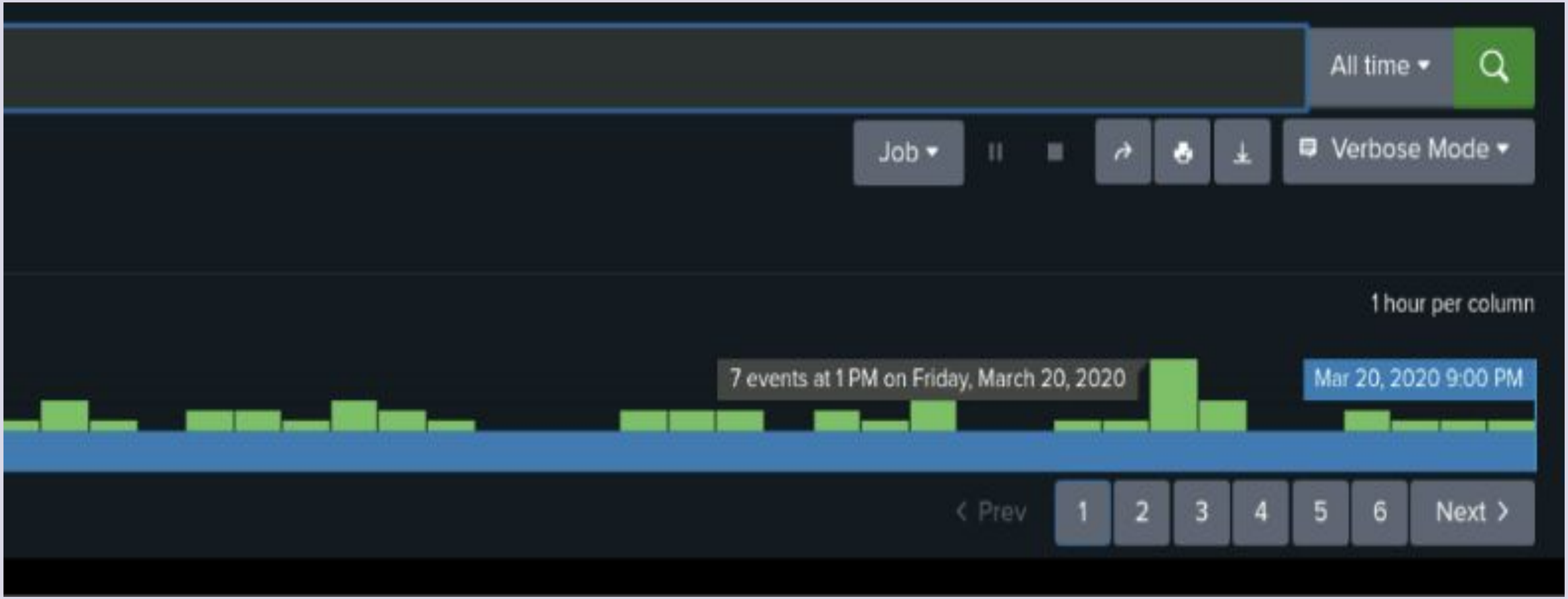
# Alerts—Apache

Designed the following alerts:

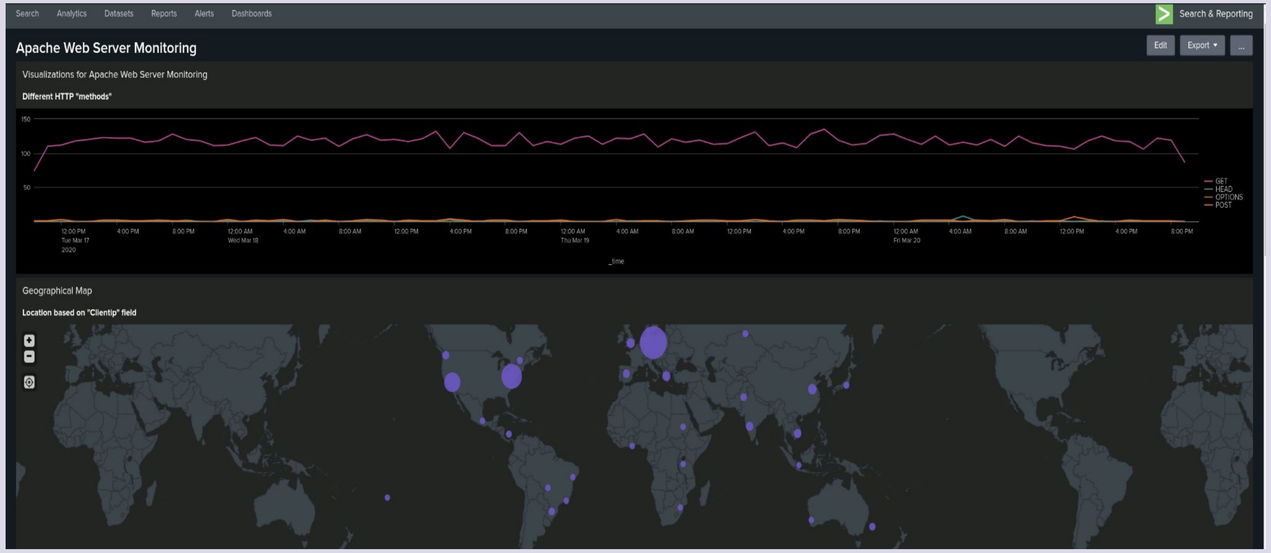
| Alert Name | Alert Description | Alert Baseline | Alert Threshold |
|---|---|---|---|
| HTTP Post Activity | The hourly count of the HTTP POST method. Baseline and threshold set as shown. | 2 | 6 |

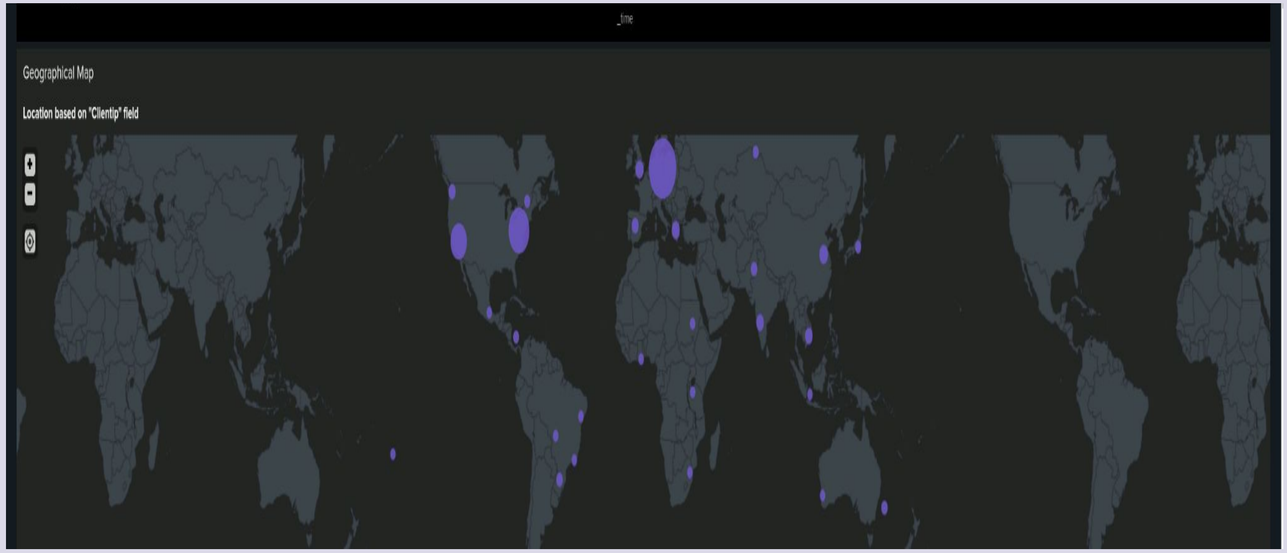
**JUSTIFICATION: Most events ranged at a low count of 2-5. Having a threshold of 6 should show any indication that there might be an attack occurring since the events never seemed to pass higher than 7 events.**

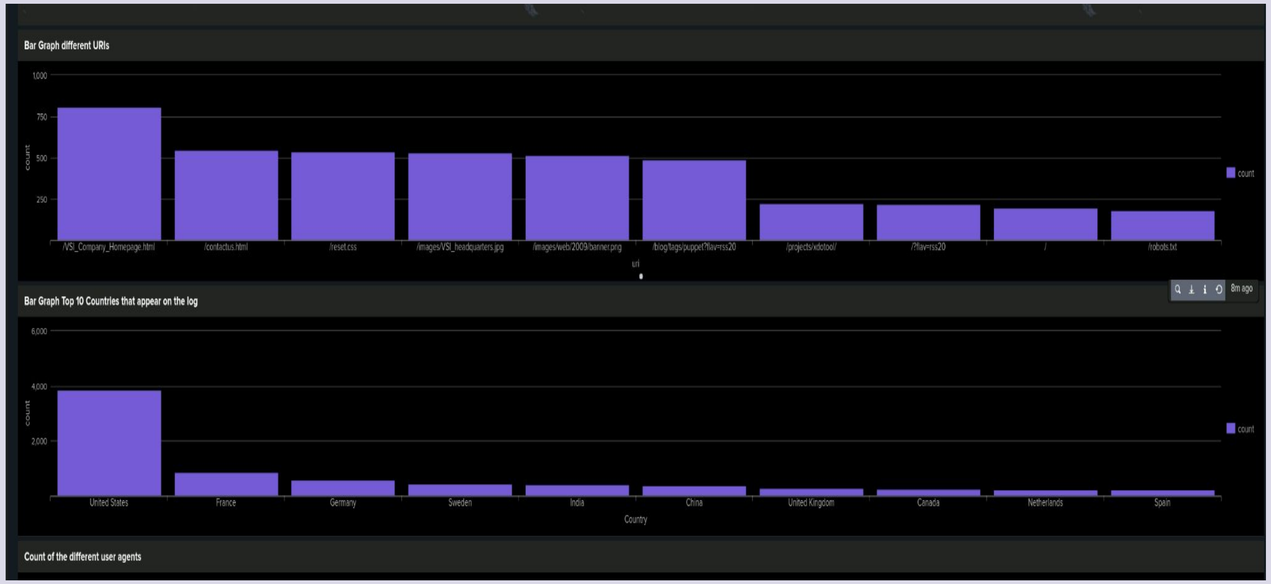
# Alerts—Apache ( HTTP POST Activity) Image

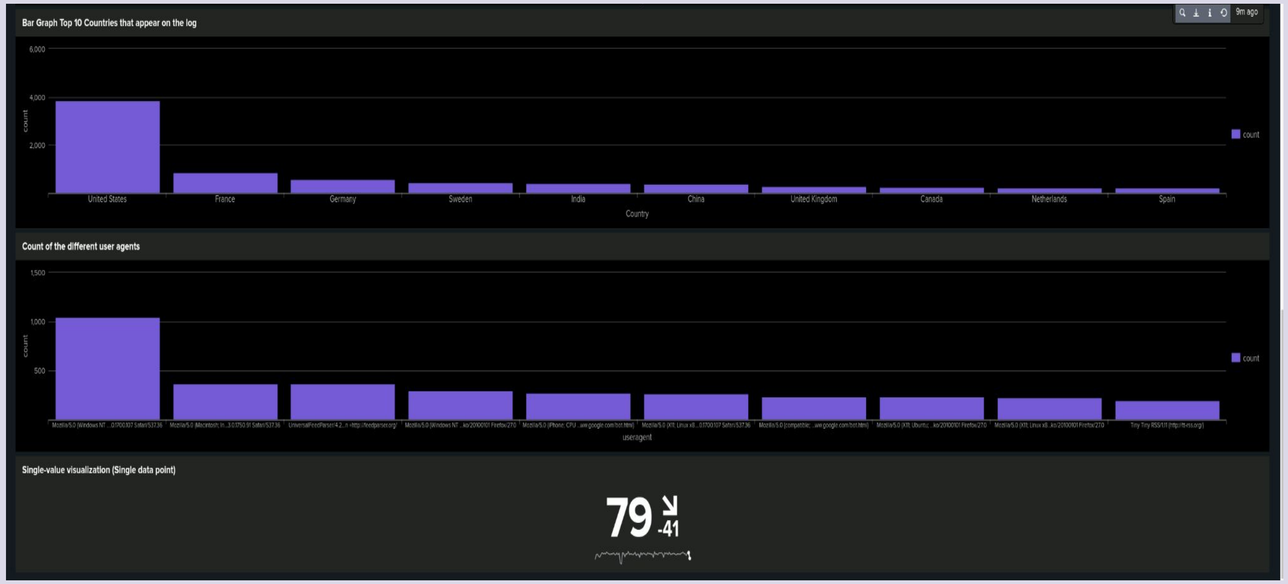


# Dashboards—Apache



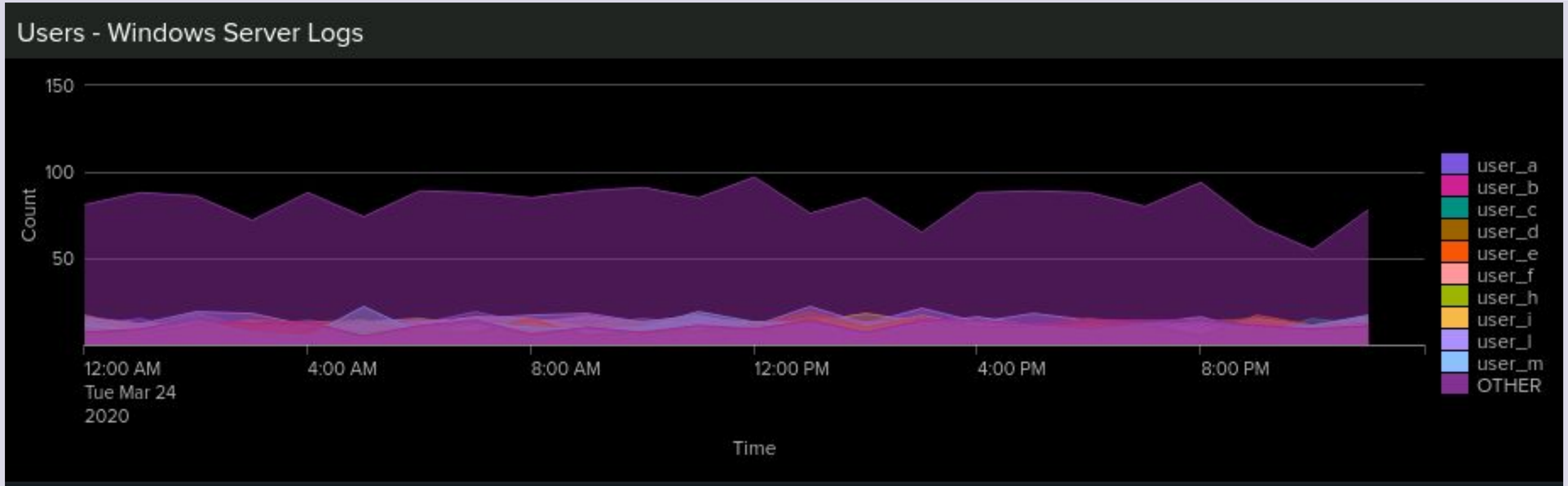



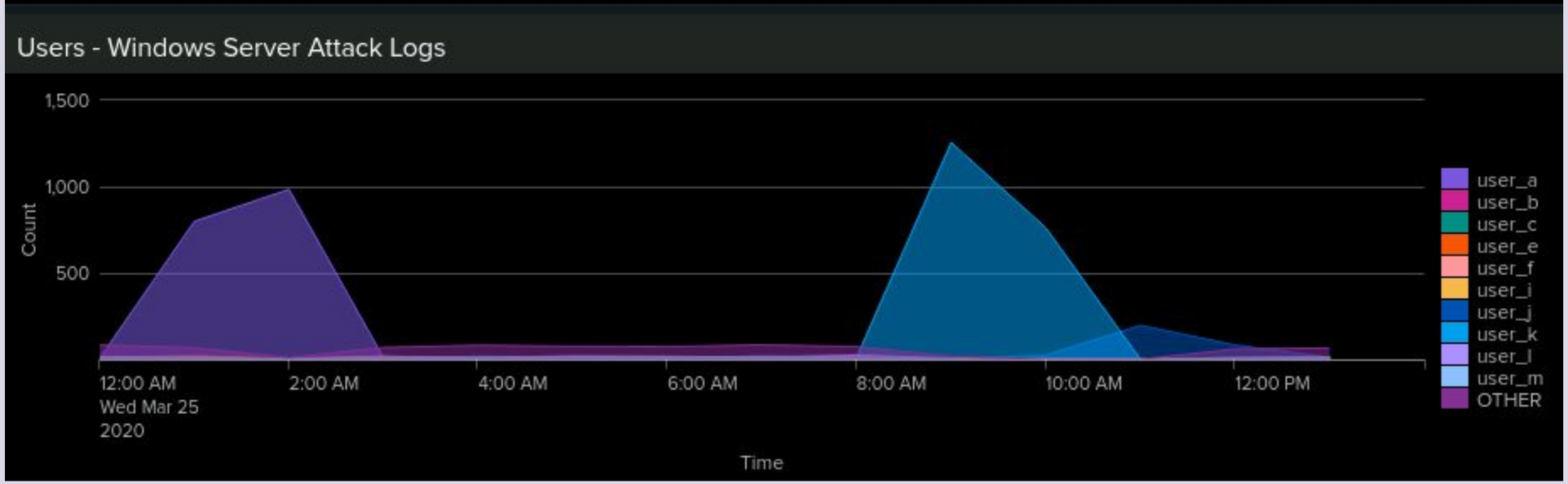



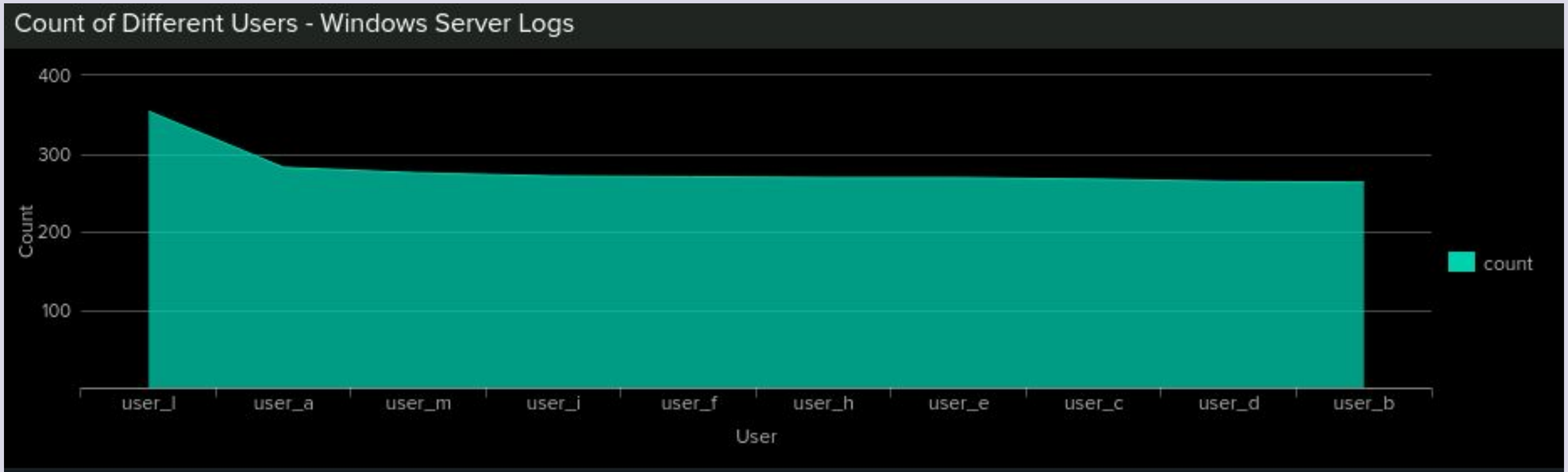
# Attack Analysis

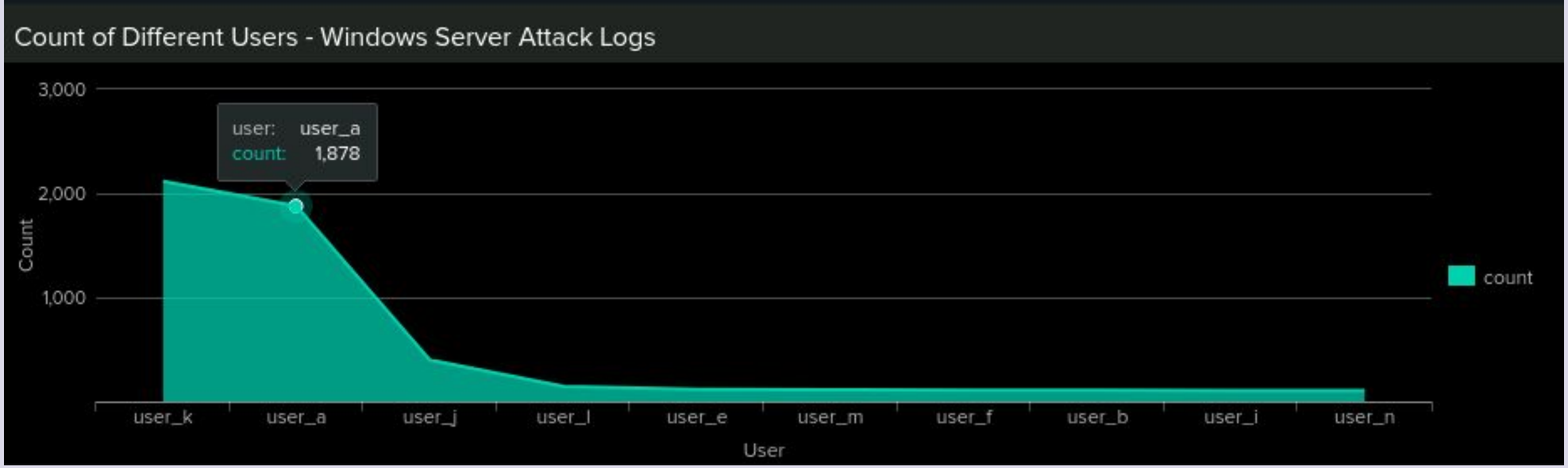
# Dashboards—Attack Logs



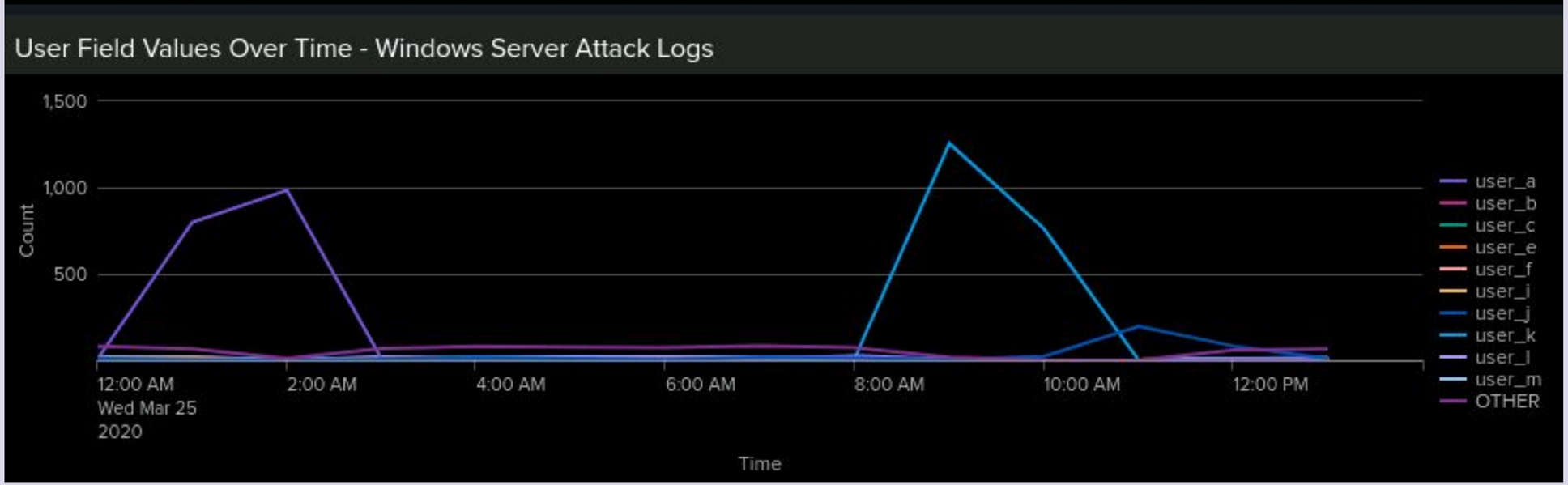



# Dashboards—Attack Logs

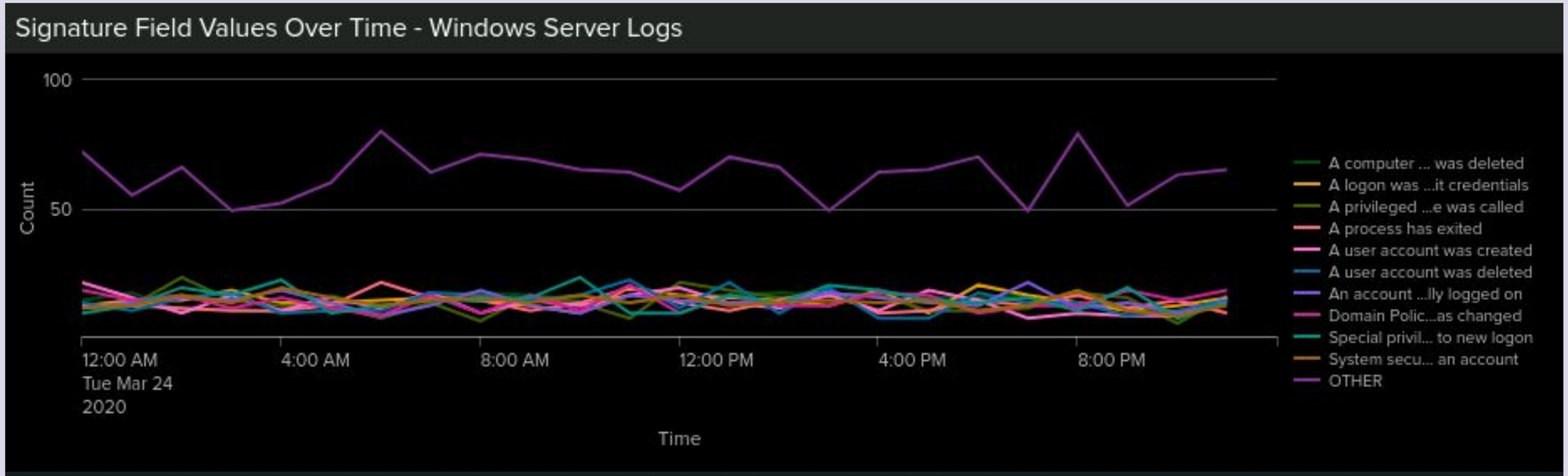


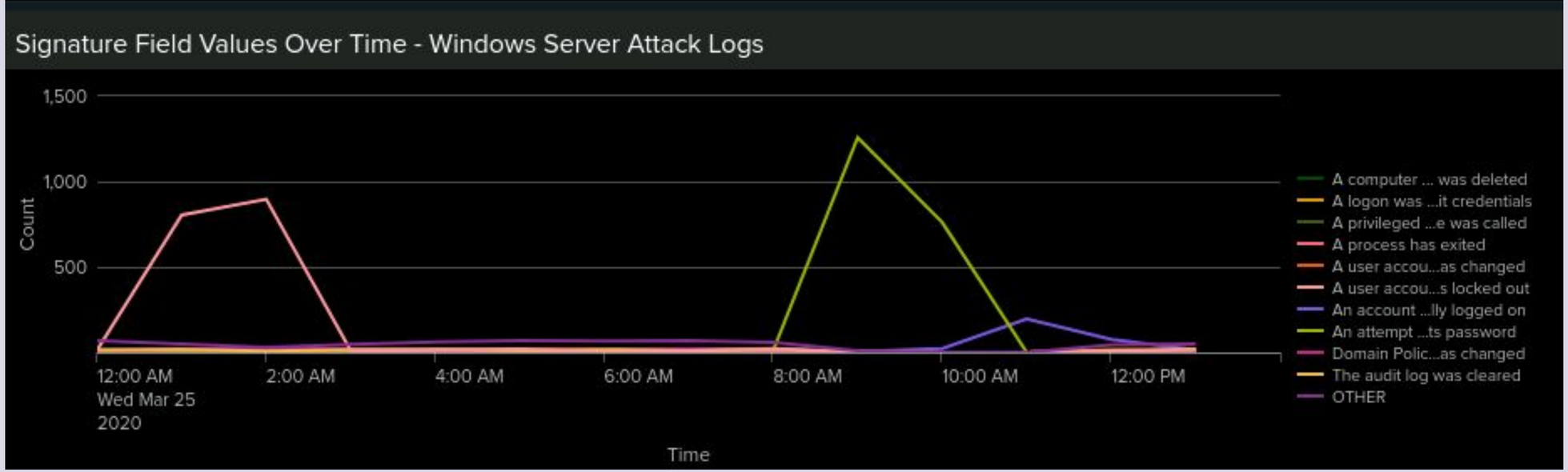



# Dashboards—Attack Logs

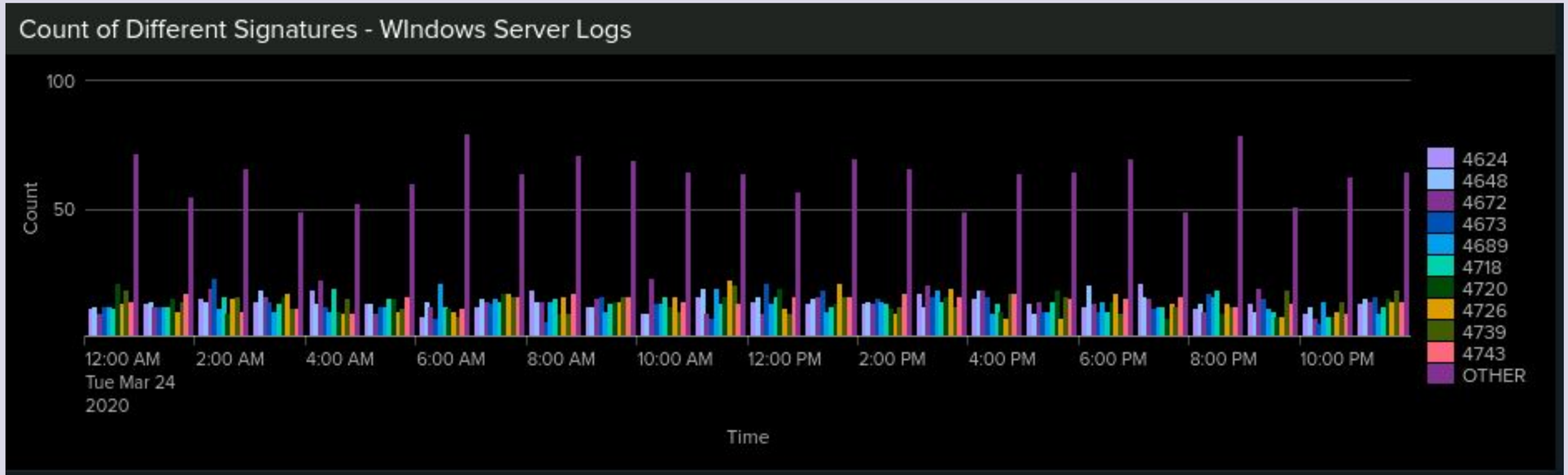


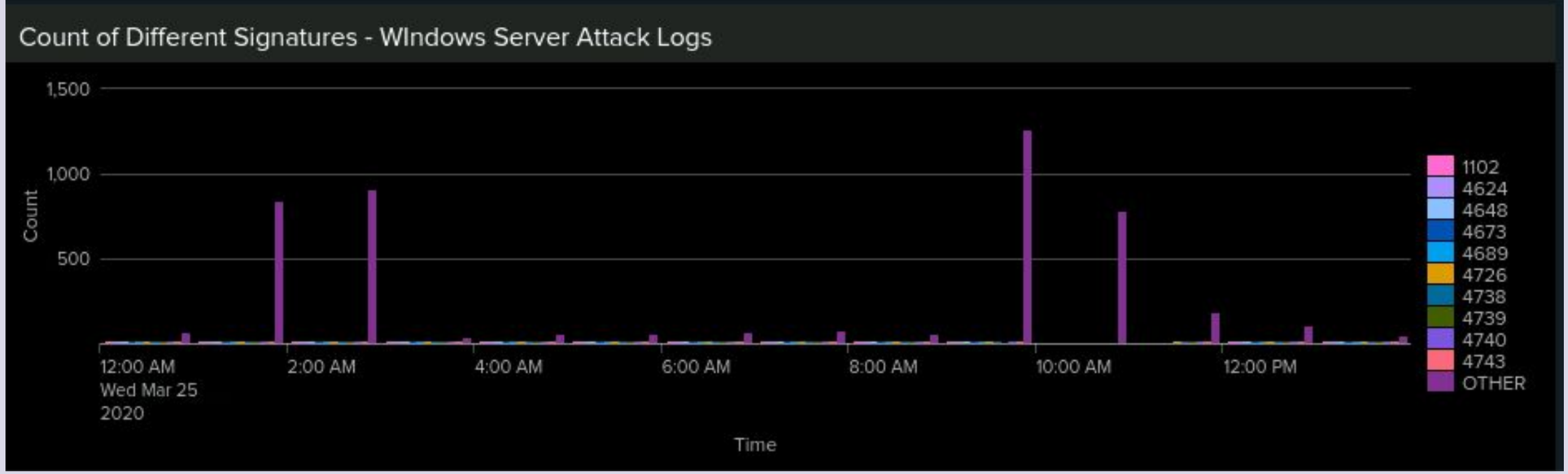
# Dashboards—Attack Logs



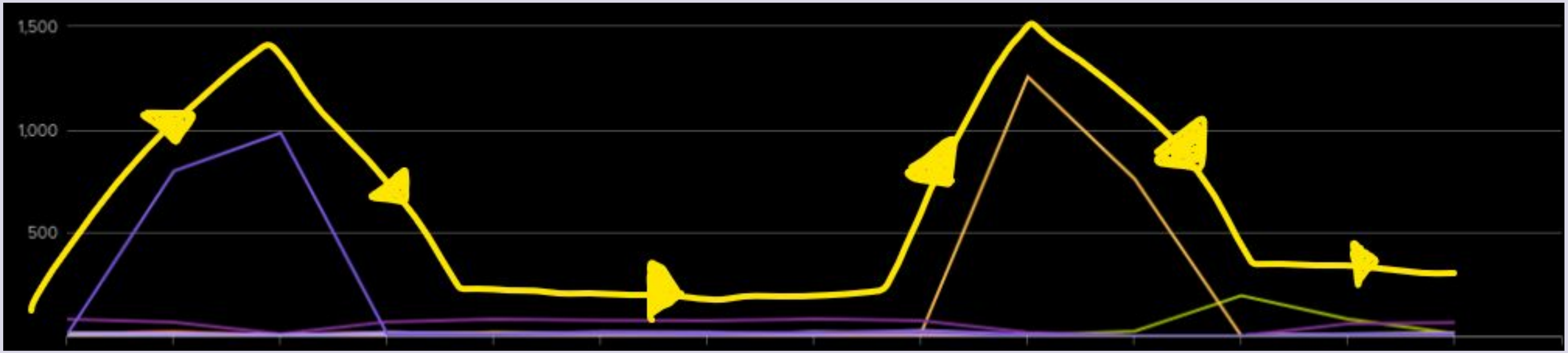
# Dashboards—Attack Logs



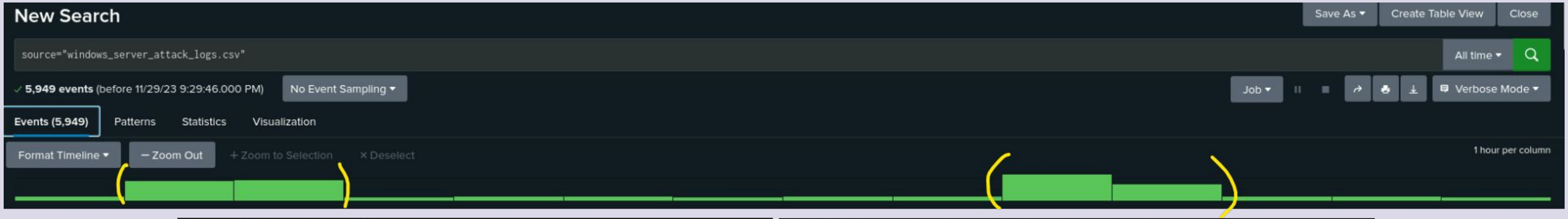



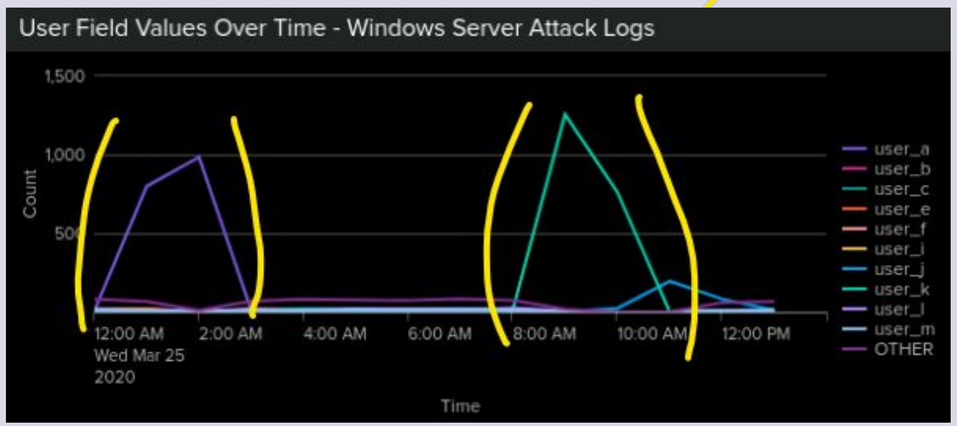
# Attack Summary—Windows

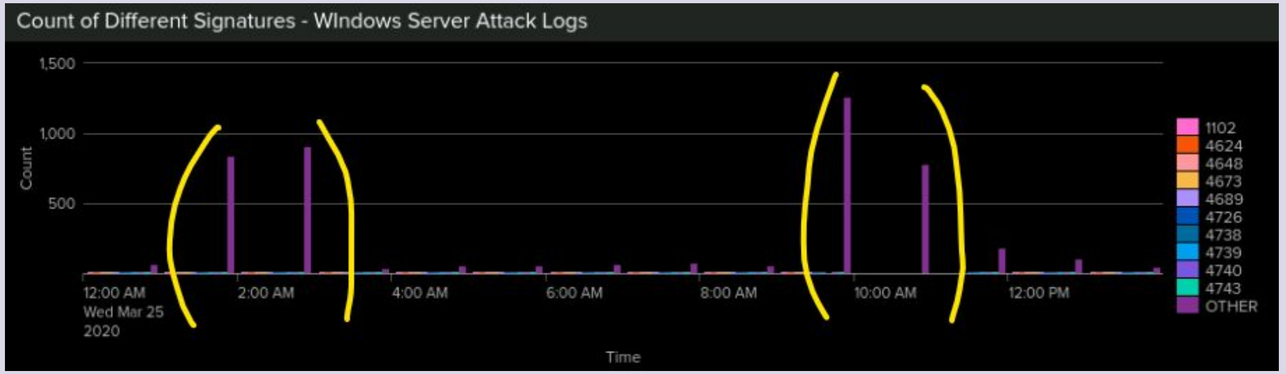
There was a distinct correlation between the attack data points where two spikes in abnormal activity can be found.



# Screenshots of Attack Logs

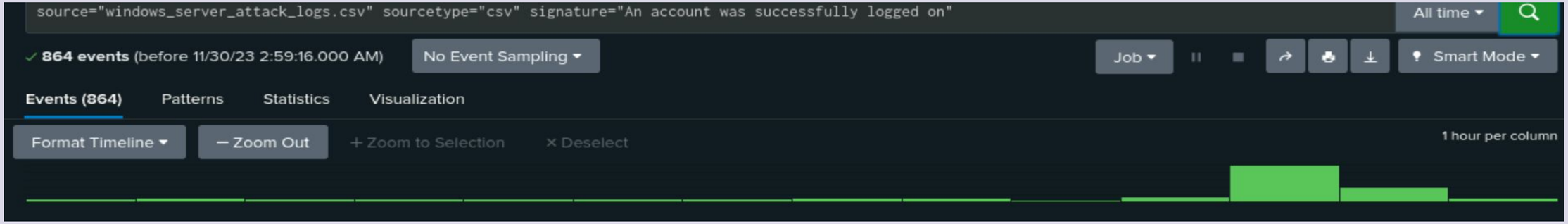






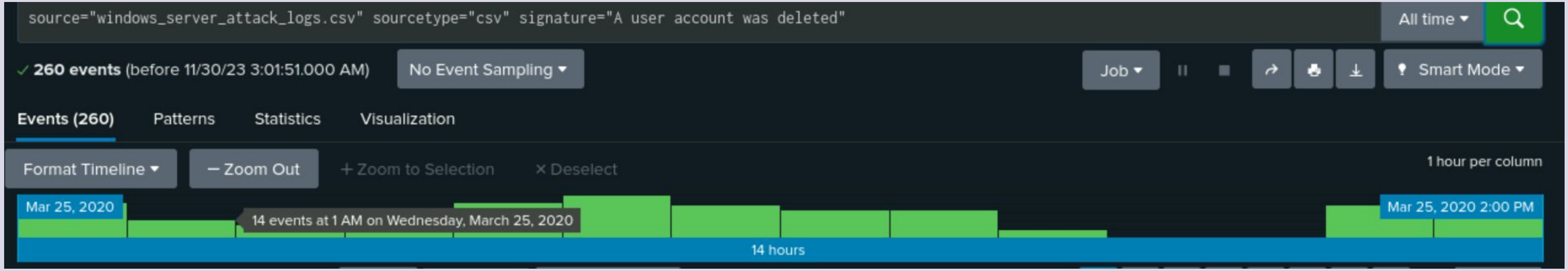
# Attack Summary—Windows

Summarize your findings from your alerts when analyzing the attack logs. Were the thresholds correct?



# Attack Summary—Windows

Summarize your findings from your alerts when analyzing the attack logs. Were the thresholds correct?



# Attack Summary—Apache

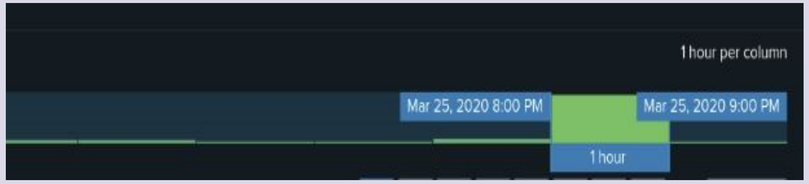
Summarize your findings from your reports when analyzing the attack logs.

- The reports showed GET method activity decreased by 28.3% while POST method activity increased by 28.3%
- There was no suspicious activity in referrer domains during the attack
- The changes in HTTP response codes showed to be on the “404” code, which demonstrated an increase from 2% to 15%.

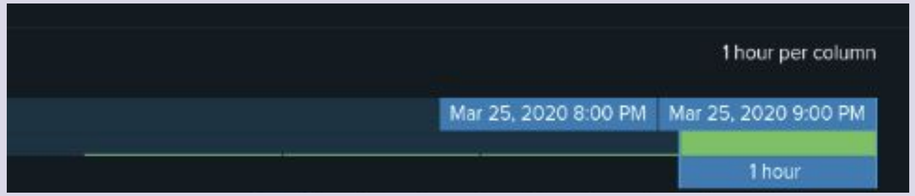
# Attack Summary—Apache

Summarize your findings from your alerts when analyzing the attack logs. Were the thresholds correct?

- Suspicious international activity was detected on Wednesday March 25, 2020 at 8:00PM in Ukraine




  - 937 events
  - Since it is above the threshold, we would receive an alert. Threshold should stay the same.
- Suspicious Volume of HTTP POST activity occurred March 25, 2020 at 8:00pm.
  - 1,296 events
  - Threshold could stay the same or changes to 4 since most events ranged from 1-2.

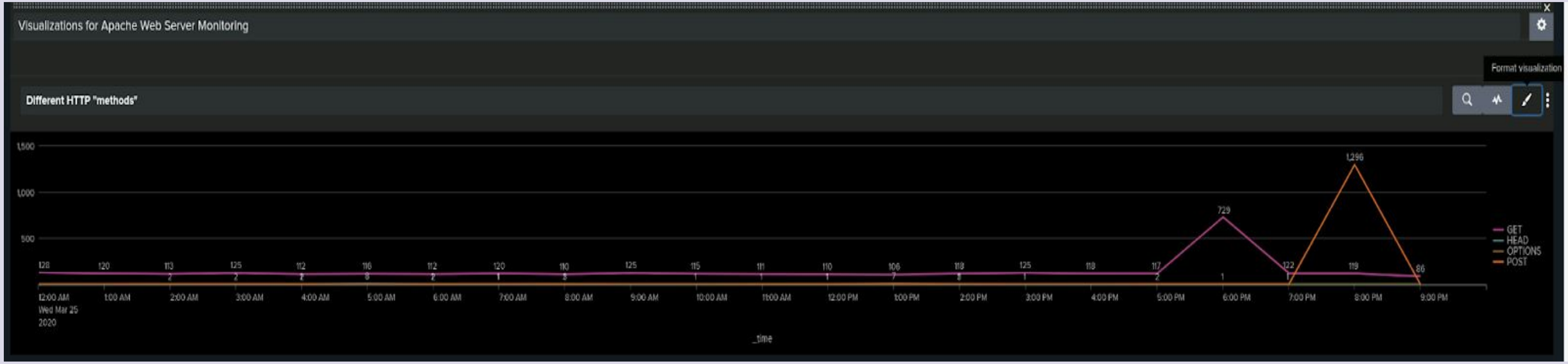


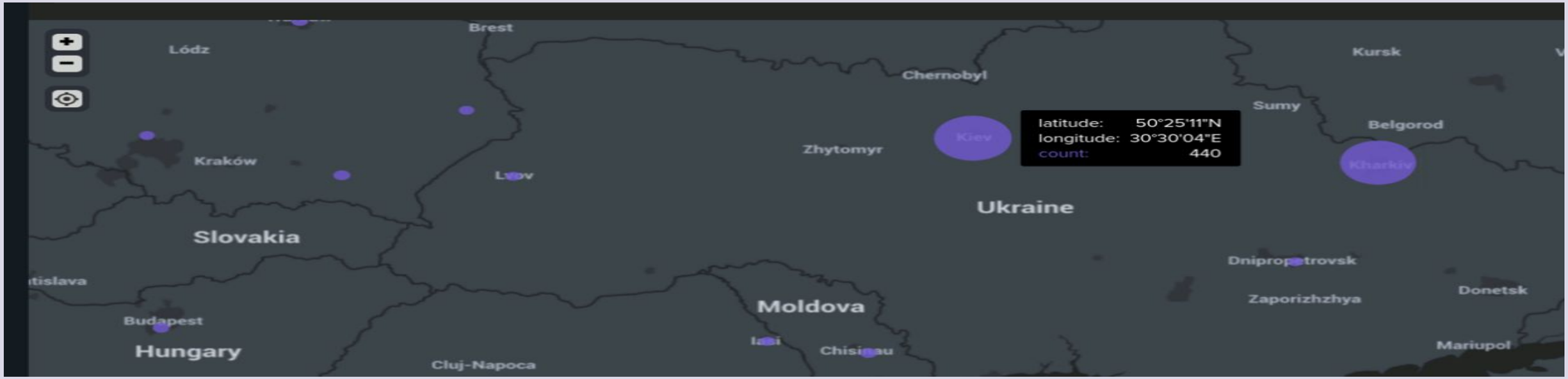
# Attack Summary—Apache

Summarize your findings from your dashboards when analyzing the attack logs.

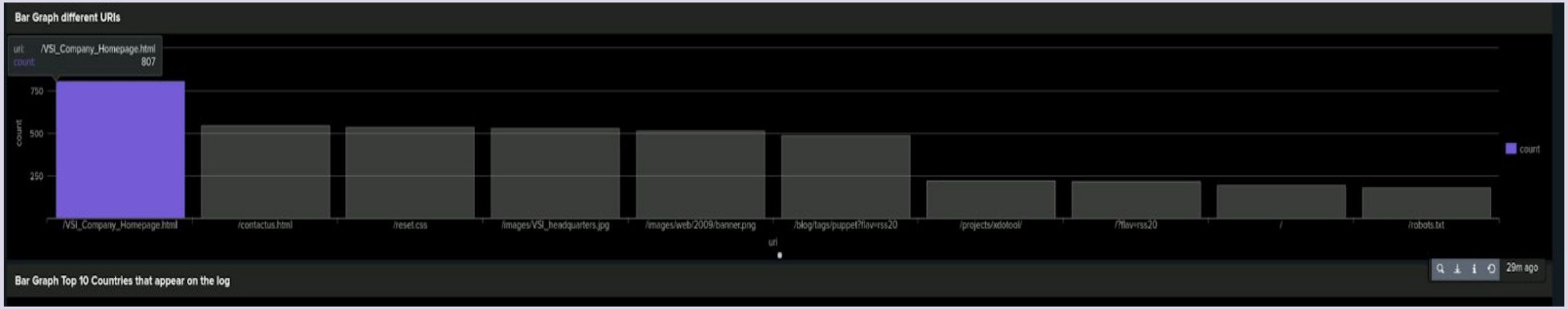
- Time Chart of HTTP Methods indicated suspicious activity:
  - GET activity increased to 729 events (5:00pm-7:00pm March 25, 2020)
  - POST activity increased to 1,296 events (7:00pm-9:00pm March 25, 2020)
- Cluster Map revealed suspicious activity from cities such as Ukraine
  - The city of Kiev (440 counts)
  - The city of Kharkiv (432 counts)
- URI data showed URI /VSI_Account_logon.php was hit the most with 1, 323 events
  - Indication of a brute force attack on the VSI login page

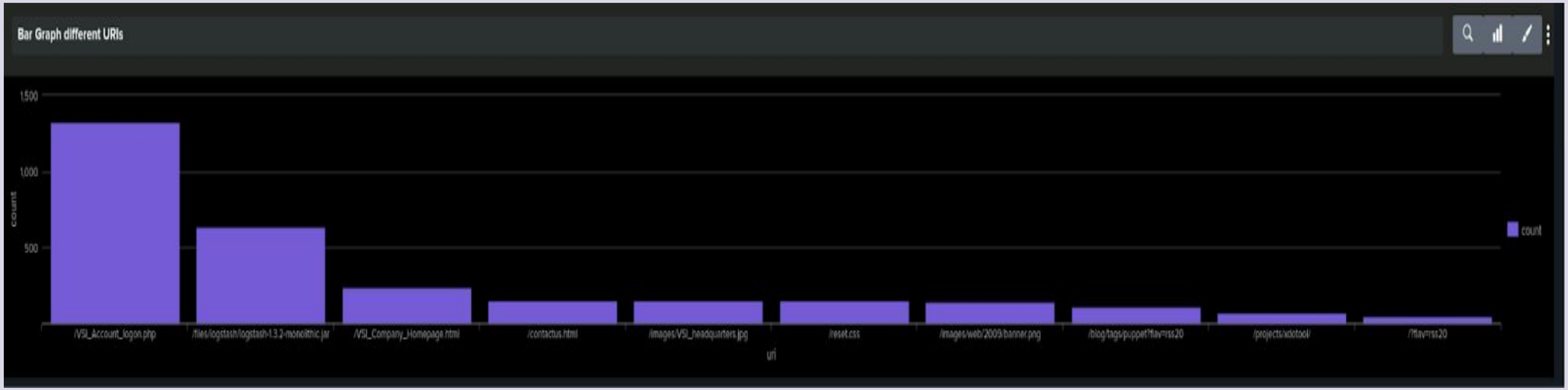
# Screenshots of Attack Logs



# Screenshots of Attack Logs



# Summary and Future Mitigations

# Project 3 Summary

- What were your overall findings from the attack that took place?
  - Our overall findings showed that on March 25, 2020 VSI had multiple attacks on their Windows and Apache servers. The attacks appear to be Brute Force attacks and guessing passwords from different countries around the world.
- To protect VSI from future attacks, what future mitigations would you recommend?
  - Using strong complex passwords, Two-Factor Authentication (2FA). Implementing CAPTCHA
  - Monitor user accounts and multiple failed login activity
  - Accounts locked  (temporarily lock user accounts after certain amount of failed attempts)